

Text Preprocessing (for NLP)

I. Steps Taken for Each Preprocessing Activity

1. Stopword Removal:

- We used NLTK's built-in English stopwords list to remove commonly used words like is, the, and, and to, which typically don't add much meaning in text analysis.
- In addition to the default list, we also removed some custom stopwords such as book, story, surprise, learn, novel, and author. These were frequently occurring in the reviews but didn't contribute significant insight in our context.
- This step helped in reducing noise and simplifying the data, allowing the analysis to focus on the more meaningful and distinctive words within each review.

2. Special Character Removal:

- Applied regular expressions — specifically `re.sub(r'^a-z0-9\s]', '', text)` — to clean the text.
- Removed unnecessary punctuation, symbols, emojis, and formatting characters that could interfere with analysis.
- Ensured that only clean and readable text remained for NLP tasks.
- Focused on retaining meaningful textual content while eliminating noise from the data.

3. Tokenization:

- Used `nlk.word_tokenize()` to split the cleaned review text into individual words or tokens.
- The resulting tokens were stored as a list format for further processing, but could also be rejoined into strings for modeling or display.

4. Case Normalization:

- Converted all review text to lowercase using `.lower()` to ensure consistency across the dataset.
- This helps avoid duplication of tokens that differ only by case (e.g., *Book* vs *book*).

- Lowercasing also reduces the overall vocabulary size, which improves text matching and model efficiency.
- Overall, this step ensures uniform text processing and leads to better feature representation for NLP tasks.

II. Sample Text: Before and After Preprocessing

a) **Original Review Text:**

I did not expect this type of book to be in library was pleased to find it price was right.

b) **After Case Normalization (lowercasing):**

i did not expect this type of book to be in library was pleased to find it price was right.

c) **After Special Character Removal:** (No special characters to remove in this case — text remains the same).

i did not expect this type of book to be in library was pleased to find it price was right.

d) **After Tokenization:**

['i', 'did', 'not', 'expect', 'this', 'type', 'of', 'book', 'to', 'be', 'in', 'library', 'was', 'pleased', 'to', 'find', 'it', 'price', 'was', 'right']

e) **After Stopword & Custom Word Removal:**

['expect', 'type', 'library', 'pleased', 'find', 'price', 'right']

III. Challenges Encountered and Solutions Applied

Empty Text After Cleaning

- **Challenge:** Some reviews became empty or NaN after stopword removal.
- **Solution:** These rows were filled with the placeholder "No Review" to maintain data integrity and avoid issues in further processing.